# From a step test to lactate thresholds



## Record the test

One row per stage, the lactate taken in the last 30 s. The resting sample is the row with intensity 0.

| intensity | lactate | heart_rate | rpe |
|---|---|---|---|
| 0 | 0.93 | 96 | 6 |
| 50 | 0.98 | 114 | 7 |
| 75 | 1.23 | 134 | 8 |
| … | … | … | … |
| 191 | 8.64 | 198 | 19 |

| | |
|---|---|
| `recording_sheet()` | A printable sheet, a web form and a CSV template with these columns. |
| `read_recording()` | Reads the .xlsx of the form: stages, recovery, the athlete. |
| `demo_data` | The test above; `lactate_tests` has 15 more, four sports. |

### A last stage not completed

Enter its time-weighted intensity: the last completed stage plus the increment × seconds done / seconds of a stage. Or leave it out with `max_intensity`.

### Make tests comparable

Same diet and training in the 24 to 48 h before, same time of day, stages of 3 min or more, two or three stages below the first rise, the same sampling site, sample type and analyzer every time.

Sources and the strength of each point: the *Best practices* article.

### Sports and units

| sport | Intensity | Also shown |
|---|---|---|
| cycling | W | per kg with a body mass |
| running | km/h | pace, min/km |
| swimming | m/s | pace, min/100 m |
| rowing | W | 500 m split |

## Estimate every threshold

```
results <- lactate_threshold(
  demo_data, intensity, lactate, heart_rate,
  rpe_column = rpe, sport = "cycling"
)
```

| Method | What it finds |
|---|---|
| Log-log | Where two lines through log lactate and log intensity cross |
| OBLA 2.0–4.0 | A fixed lactate, 2 to 4 mmol/L |
| Bsln+ 0.5–1.5 | 0.5 to 1.5 mmol/L above the resting value |
| $D_{max}$ family | The point furthest below a line to the last stage |
| LTP1, LTP2 | The turn points of three straight segments |
| $LT_{ratio}$, IAT | The lowest lactate / intensity; 1.5 mmol/L above it |
| Stegmann | A tangent from recovery to the curve |

### Useful arguments

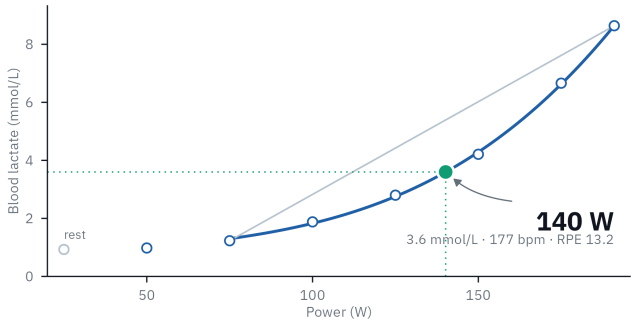
| | |
|---|---|
| `method` | Some methods only, e.g. `c("Dmax", "LTP")` |
| `fit` | The curve: polynomial (3rd or 4th degree) or B-spline |
| `sport` | cycling, rowing (W), running (km/h), swimming (m/s) |
| `ci = TRUE` | Bootstrap confidence intervals |
| `recovery` | Recovery samples and `stage_duration`, for Stegmann |
| `sample` | blood, plasma, sweat or interstitial fluid |
| `.by` | Many tests in one call, e.g. `.by = "athlete"` |

### Fine-tune the methods

| | |
|---|---|
| `fixed` | OBLA targets, mmol/L; default 2 to 4 |
| `baseline_plus` | Bsln+ rises, mmol/L; default 0.5, 1, 1.5 |
| `iat_rise` | The IAT rise above $LT_{ratio}$; default 1.5 |
| `dmax_anchor` | Where the $D_{max}$ lines start and end |
| `ltp_breakpoints` | 1 for a test that stopped before LTP2 |
| `max_intensity` | Leave out stages above it |

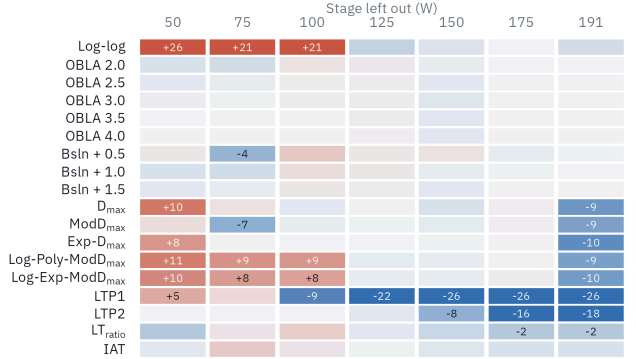
## Look, then check

```
autoplot(results)                       # every method
autoplot(results, method = "ModDmax") # one method
```




One method: the curve, its line, and the threshold at 140 W.

| | |
|---|---|
| `lactate_checks()` | A checklist of what is easy to miss: missing rest, a lactate dip, a final plateau |
| `lactate_diagnostics()` | Residuals, influential stages, how far each threshold moves without one stage |
| `lactate_fit_quality()` | Which curve predicts the stages best |




Leave one stage out: red, the threshold rises without that stage; blue, it falls; dark, it rests on it.



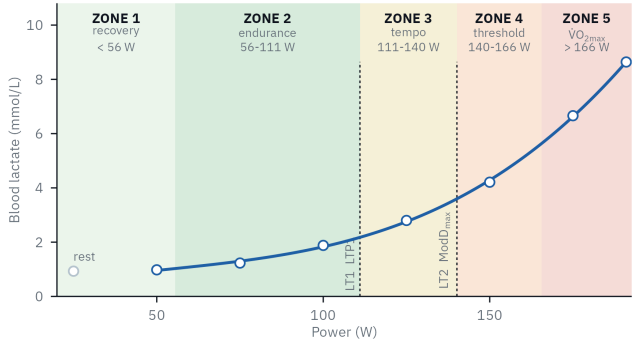
lactater 1.0.0, MIT licence

# Using the thresholds



## Zones and training

```
zones <- lactate_zones(results, lt1 = "LTP1",
                       lt2 = "ModDmax", model = 5)
```




LT1 and LT2 split the intensities into moderate, heavy and severe; five zones split the outer two.

| | |
|---|---|
| `model` | 3 zones (the domains) or 5 |
| `split` | Where zones 1\|2 and 4\|5 sit: `"equal"`, `"seiler"`, or your own |
| `training_load()` | Time in each zone from training files, by power or heart rate |

```
training_load(sessions, zones, column = "power",
              measure = "intensity", interval = 1)
```

### Choosing LT1 and LT2

| | Methods with some support |
|---|---|
| LT1 | Log-log, Bsln+ 0.5: moderate to heavy domain |
| LT2 | Bsln+ 1.5 (3-min stages), Log-Poly-ModD$_{max}$ (4-min stages): close to the maximal lactate steady state |

Jamnick et al. 2018 and 2020; few methods have evidence for the heavy to severe boundary. Report the pair you chose.

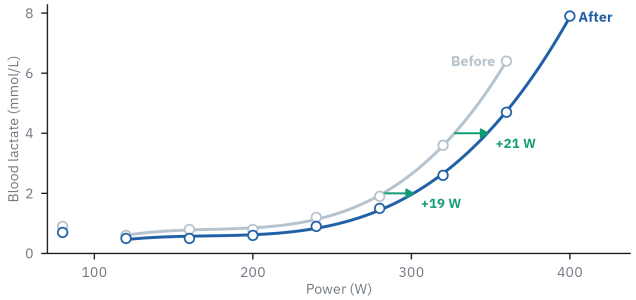
### Heart rate and RPE

Every zone gets heart rate and RPE ranges when they were recorded. Heart rate lags behind intensity in short intervals, so heart-rate zones suit steady efforts.

## Predict and compare

```
predict_performance(results)
own <- calibrate_performance(lab, "ltp2", "time",
  method = "LTP2", sport = "running", distance =
10000)
predict_performance(results, equation = own)
```

Race times (running) and 45-min time-trial power (cycling), each with its 95% range, from published equations or your own athletes.




| | |
|---|---|
| `compare_tests()` | Is a change larger than the noise of the method? |
| `curve_shift()` | How far the whole curve moved |
| `curve_shape()` | Did it shift, or change shape? |
| `lactate_trend()` | Where a series of tests is heading |
| `relative_thresholds()` | Per kg, % of peak intensity and heart rate |
| `reference_percentiles()` | Percentile among people of the same age and sex (cycling, general population) |

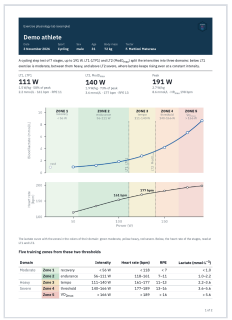
### What the predictions rest on

| Sport | From | Source |
|---|---|---|
| Running | IAT: 5 km to marathon; OBLA 4.0: marathon | Röcker 1998; Sjödin 1981 |
| Cycling | Six thresholds: 45-min time trial | Heuberger 2018, refitted |
| Rowing, swimming | Your own equation | `calibrate_performance()` |

## Report and style

```
report_start(results, "LTP1", "ModDmax") |>
  report_header(name = "Ana", lab = "Our lab",
                logo = "logo.png") |>
  report_summary() |>
  report_zones() |>
  report_save("ana.html", pdf = TRUE)
```

| | |
|---|---|
| `test_report()` | The default two-page report in one call; 2-5 tests and their `dates` for the progress |
| `report_*()` | Header, summary, zones, prediction, checks, test, methods, your own text |
| `lactate_report()` | Every threshold as a gt table for a paper |
| `gt_lactater()` | The same look for your own gt tables |



The first page of a report: the two thresholds, what they mean, the zones.

### Plots

| | |
|---|---|
| `theme_lactater()` | The look of lactater's plots, for your own |
| `lactater.style` | `options(lactater.style = "night")`, or `"journal"` for print |
| `lactater.font` | `options(lactater.font = "IBM Plex Sans")` |

### Beyond the step test

| | |
|---|---|
| `vlamax()` | The maximal lactate accumulation rate ($\dot{V}La_{max}$) from a sprint |

### Learn more

Every method follows its original paper; the articles explain each one, its evidence, and how to read the checks.



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